A Novel Method for Mapping Moso Bamboo Forests Using Remote Sensing Data With the Consideration of Phenological Status

Xuying Huang, Weimin Ju, Zhanghua Xu, and Jing Li

***Abstract*— Precisely delineating the distribution of moso bamboo forests is critical for forestry management and regional carbon cycle research. The unique phonological characteristics (i.e., on- and off-year phenomenon) of bamboo impose difficulties in bamboo identification. This study aims to develop a new algorithm for mapping bamboo distribution using remote sensing data with the consideration of bamboo phenological characteristics. Three optical indices were proposed based on canopy reflectance retrieved from Sentinel-2 and field inventory data, including modified bamboo index (MBI), bamboo phenological characteristic index (BPCI), and BPCI 2 (BPCI-2). The collaboration of these three indices with the recursive feature elimination (RFE) and extreme gradient boosting (XGBoost) methods can precisely map bamboo distribution and its phenological status. The model based on MBI, BPCI, and BPCI-2 outperformed the model driven by the existing bamboo extracting indices, i.e., bamboo index (BI), yearly change bamboo index (YCBI), and monthly change bamboo index (MCBI), increasing in overall accuracy (OA) by about 1.5%. Additionally, the proposed indices were calculated using the data synthesized from Sentinel-1 synthetic aperture radar (SAR) imageries by the cycle-consistent adversarial network (CycleGAN) method under the condition without cloudy-free Sentinel-2 data available to fill the time series data gaps. The performance of the model based on augmented data improved notably in comparison with the model driven only by indices from original optical images, with the identification accuracy for on- and off-year bamboo samples over 96%. The generated moso bamboo distribution map aligns well with forestry inventory data in terms of both area and spatial distribution. The proposed indices are less sensitive to terrain than the existing bamboo extracting indices. This merit is valuable for better mapping bamboo forests, which are mostly distributed in mountainous areas.**

***Index Terms*— Forest, generative adversarial networks (GANs), machine learning, moso bamboo, remote sensing, spectral.**

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I. Introduction

BAMBOO is a fast-growing gramineous plant that mainly grows in tropical and subtropical regions of Asia, Africa, and Latin America. China has numerous bamboo resources, i.e., the coverage of bamboo forests is about 6.41 million ha, accounting for about 20% of the global total [1]. The annual output value of China's bamboo industry in 2020 was about 300 billion RMB Yuan and provided more than 10 million jobs [2]. Moso bamboo (*Phyllostachys pubescens*) accounts for 72.96% of China's bamboo forests area and its product comprises most of the transaction content in the bamboo industry [3]. Bamboo forest also plays an important role in mitigating global climate change [4]. Previous studies indicated that the mean aboveground carbon sequestration of moso bamboo (i.e., 8.13 $\pm$ 2.15 Mg/ha) per year is 2.43$\times$ that of Chinese fir (i.e., 3.35 $\pm$ 2.02 Mg/ha) [5]. The carbon storage of bamboo products is about 1.67$\times$ wood products [6]. Therefore, bamboo forests could make an important contribution to the realization of carbon neutrality strategy. However, the over management of bamboo forests will cause a series of environmental problems [7], e.g., soil degradation, biodiversity decline, and its invasive growth pattern impose a great threat to the native tree species [8]. Therefore, the spatiotemporal dynamic pattern and up-to-date distribution information of bamboo forests are required for better development of the bamboo industry.

Currently, the bamboo forest distribution information is mainly obtained from manual field inventories. However, the inventory data only provides the aggregation of forest species information at the stand scale and requires expensive costs (e.g., time, labor, and economics). In heavily forested areas laboriously accessible, the inventory work may even threaten the personal safety of investigators. Therefore, the remote sensing-based bamboo composition map is demanded for effective forestry management. The application of remote sensing in forestry resources monitoring has achieved much success at various spatiotemporal scales. High-resolution

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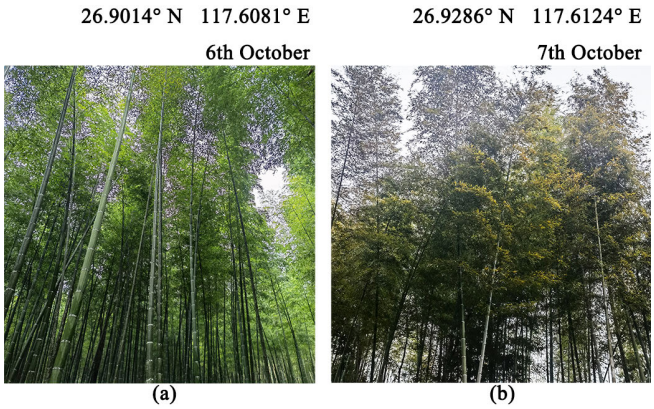


Fig. 1. Morphological difference between (a) on-year and (b) off-year bamboo.

remote sensing imagery, e.g., aerial sketch data, unmanned aerial vehicle image, and submeter remote sensing satellite data, has been successfully applied to map tree species distribution [9], [10]. However, the application of such remote sensing data is usually restrained in small areas because of the expensive price [11]. Freely available moderate satellite remote sensing imagery, e.g., Landsat [12], [13], and Sentinel [14], [15], enables it possible to map forest species at a large scale in an economical way and to track the spatiotemporal change patterns of forests.

Since the leaf physicochemical status (e.g., chlorophyll, moisture, and shape) varies among various tree species, their spectrum is certainly different. Meanwhile, the canopy structure of specific tree species has unique morphological characteristics. Differences in sunlight distribution within the canopy lead to reflectance variations among tree species with different crown morphologies. The commonly used remote sensing features for bamboo extraction include principal component analysis variables [16], the specific band combination [17], [18], and the vegetation index [19], [20], [21]. Many studies used time series of remote sensing data to capture the seasonal variation information of moso bamboo to improve the classification and have achieved much success [20], [22]. The appropriate selection of time phase images is critical for extracting bamboo. Wei et al. [23] analyzed the seasonal spectral characteristics of bamboo forests using MODIS image data and found out that early or middle autumn is the best period for extracting bamboo forests, followed by summer. While Li et al. [24] declared that on Sentinel-2 and Landsat-8 imageries the spectral difference of moso bamboo from other forests maximized from April to May. It is noteworthy that most previous studies on bamboo forest mapping did not consider bamboo phenological characteristics, i.e., on- and off-year. The morphological features of moso bamboo differed greatly between on- and off-year periods (Fig. 1). Without proper consideration of difference in on- and off-year bamboo, the accuracy of mapping bamboo forests and the derivative products (e.g., leaf area index and gross primary productivity) will be significantly affected [25], [26].

Generally, the shoots incubation period from the end of leaf expansion to the production of the new shoot in the next year is called on-year. The leaf changing period from the old leaf discoloration and shedding to the sprout of new leaves in the next year is called off-year [27]. There is considerable spectral difference between on- and off-year bamboo [28]. For the recognition of bamboo phenological status, Chen et al. [29] constructed the monthly change bamboo index (MCBI) based on the spectral differences of on- and off-year bamboo forests in red-edge and near-infrared bands between April and May. Li et al. [24] proposed the yearly change bamboo index (YCBI) according to the interannual spectral difference of moso bamboo in May. They found that the combination of YCBI and MCBI can not only be used to extract distribution information on moso bamboo forests but also can be used to distinguish the on- and off-year bamboo regions. However, moso bamboo mostly grows in mountainous areas. The impact of terrain on the extraction of bamboo forests has not yet been investigated. Additionally, clouds are abundant in many tropical and subtropical regions. It is difficult to acquire cloud-free optical images during the key time phase for bamboo extraction.

Many image compositing methods have been proposed to overcome the limitation of optical data acquisition, e.g., radial basis convolution filters [30] and enhanced spatial and temporal adaptive reflectance fusion model [31]. But the quality of the composited images is not guaranteed if the data gap is too long. Recently, deep learning has been widely applied in the field of remote sensing [32], [33], [34]. The generative adversarial networks (GANs) have been successfully used for remote sensing data synthesis [35], [36]. GANs involve two critical components: the generator and the discriminator. The generator takes certain inputs and generates synthetic images resembling real images to deceive the discriminator. The task of the discriminator is to learn the features of real data and identify the differences between real images and synthetic images. This competitive process continues through iterations until the generated images become indistinguishable from real ones [37]. CycleGAN is a popular algorithm for performing cross-modal image processing. It has the ability to achieve domain-to-domain image translation without the need for paired training data [38]. The synthetic aperture radar (SAR) is able to provide equidistant time series data because of the ability of active microwave to penetrate the clouds. Combining SAR data and the CycleGAN method to generate missing optical images is a feasible approach for data augmentation [39], [40].

Identifying tree species in complex forest areas is challenged by the issue of the same substance having different spectra or different substances having the same spectrum. The critical aspect of the task of extracting bamboo forests lies in the selection of appropriate features. Therefore, this study is going to develop a method for extracting on- and off-year moso bamboo forest distribution from remote sensing data. The specific objectives include the following: 1) to analyze the spectral characteristics of on- and off-year bamboo and construct novel indices usable for mapping bamboo forests in mountainous regions; 2) to assess the applicability of proposed indices in mapping bamboo forests; and 3) to investigate whether the synthesis of optical and SAR data implemented

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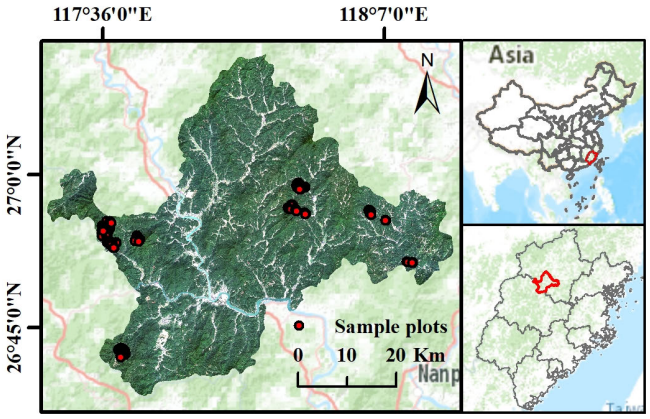


Fig. 2. Location of Shunchang County, Nanping City, Fujian Province, China. The red dots are the location of field sample plots.

using the CycleGAN method was able to improve the mapping accuracy of bamboo forests.

II. Materials and Methods

A. Study Area and Data Collection

Our study was conducted in Shunchang County, Nanping City, Fujian Province, southeast China. It spans an area of 1985 km^2 as shown in Fig. 2. The study area is mainly mountainous and hilly, with an average altitude of about 412 m. It has a subtropical marine monsoon climate with distinct dry and humid seasons (annual precipitation is nearly 1800 mm). The forest coverage here is about 80%. The local dominant species include Chinese fir (*Cunninghamia lanceolata*), Masson pine (*Pinus massoniana*), moso bamboo, and some broadleaf trees, e.g., Sweetgum (*Liquidambar formosana*) and Nanmu (*Phoebe zhennan*). Moso bamboo accounts for about 20% of the total forests' area. Therefore, the county is known as one of the "bamboo town" in China.

A field survey was conducted over coniferous, broadleaf, on-year moso bamboo, and off-year bamboo forests from 28 September to 15 October 2020. With forest inventory data provided by the local forestry bureau as the guide, we surveyed 325 sites in seven forest areas. In each site, a 30 × 30 m typical plot was set, and the dominant tree species and GPS coordinate information within each sample plot were recorded. The pest damaged condition and logging information of each bamboo forest site were also documented.

B. Remote Sensing Data Processing

As to Sentinel-1 data, the ground range detected (GRD) products in the middle of each month of 2020 were used. The preprocessing of Sentinel-1 data was conducted using the Sentinel-1 toolbox in SNAP (ESA Sentinel Application Platform v7.0), including the following steps: retrieving satellite orbit information; removing the thermal noise of images; radiometrically calibrating the original signals into backscattering coefficient in vertical–vertical (VV) and vertical–horizontal (VH) polarization; performing the range Doppler terrain correction to generate geometrically undistorted images with 10 m pixel size; and converting (i.e., logarithmic transformation) the linear scale backscattering coefficient into decibels unit (dB) as $\sigma^o vh$ and $\sigma^o vv$ [41].

Through visualization, all available standard Level-1C (L1C) top-of-atmosphere (TOA) of nearly cloud-free Sentinel-2 products in 2020 (imaging on February 21, April 16, October 23, November 15) and 2021 (imaging on February 20, November 27) were downloaded from the Copernicus Open Access Hub. The sen2cor plugins in SNAP was applied to convert the L1C product into L2A bottom-of-atmosphere (BOA) reflectance. The bands with spatial resolutions of 20 m (red-edge 1–3, NIR narrow, and SWIR bands) and 60 m (coastal aerosol and water vapor bands) were resampled into 10 m using the sen2res plugin [42].

On the Sentinel-2 images acquired on October 23, polygons around centers of individual plots were sketched based on the recorded coordinate information. In total, there are 80 polygons for coniferous (719 pixels), 81 for broadleaf (727 pixels), 84 for on-year bamboo (747 pixels), and 80 for off-year bamboo (720 pixels). These were recorded as experimental samples. Boundaries of delineated polygons were applied for other remote sensing imageries.

C. Construction of Bamboo Extraction Index

According to the spectrum of bamboo, coniferous, and broadleaf forests in winter, spring, and autumn, bamboo forest is more distinguishable from others in the shortwave infrared region (Fig. 3). The reflectance of bamboo in bands 11 and 12 of Sentinel-2 is higher than that of coniferous and broadleaf forests. Since moso bamboo usually grows in steep hilly areas, the direct application of original reflectance for tree species identification will be strongly hindered by mountain shadows. It was found that the reflectance of each tree species at about 740 nm differed marginally, so the reflectance of band 6 and a shortwave infrared band was combined to construct a modified bamboo index (MBI) in the normalized ratio form to reduce the interference from terrain, and to enhance the high reflectance of bamboo in the shortwave infrared region

$$\mathrm{MBI} = \frac{(\mathrm{Band_i} - \mathrm{Band_6})}{(\mathrm{Band_i} + \mathrm{Band_6})} \tag{1}$$

where i denotes 11 or 12, representing the SWIR-1 or SWIR-2 band in the Sentinel-2 images.

Although MBI has the potential to differentiate bamboo forests from other types of forests, it is unable to separate on- and off-year bamboo forests. It was found that the reflectance of these two bamboo forest types was notably different in the range from red to near-infrared region (Fig. 4). In February, the leaves of on-year bamboo were not completely replaced, while the mature leaves of off-year bamboo were not discolored yet. Therefore, the canopy reflectance of on-year bamboo is higher in the visible and shortwave infrared regions, and lower in the near-infrared band. In April, the leaves of the on-year bamboo gradually expanded, and its reflectance in the near-infrared region increased accordingly. At this stage, the leaves were still immature and the contents of leaf pigments, such as chlorophyll, were relatively low. As a consequence, the reflectance of on-year bamboo in the visible band is still high. In October, the leaves of on-year bamboo matured, and its

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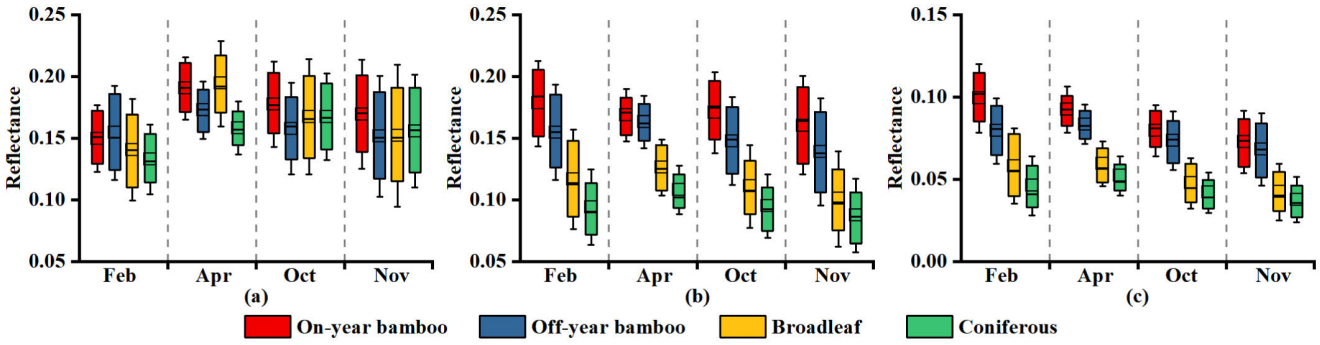


Fig. 3. Mean spectrum of different forest types in (a) band 6, (b) band 11, and (c) band 12 of Sentinel-2 images.

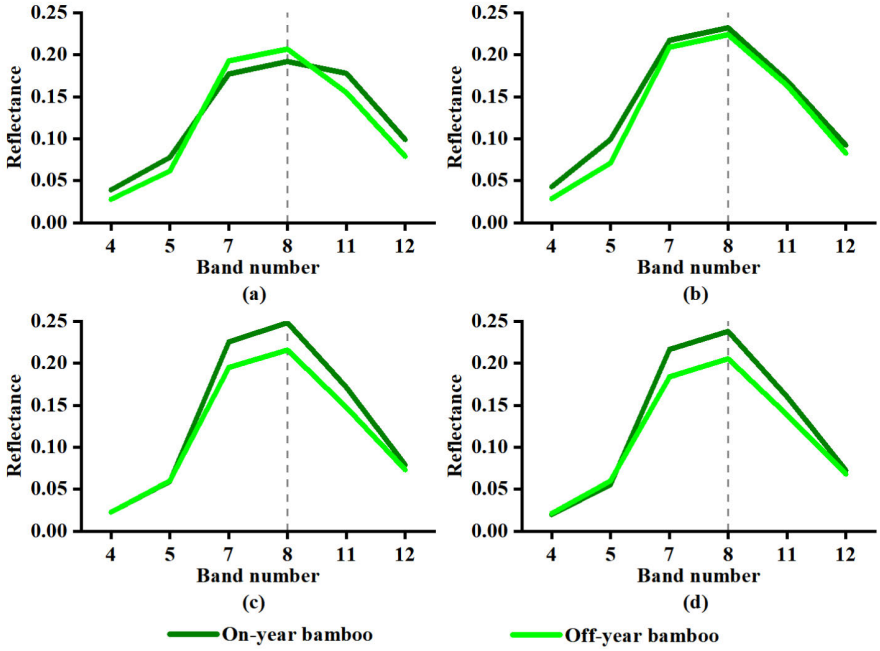


Fig. 4. Spectrum differences between on- and off-year bamboo canopy in two regions (red-NIR and NIR-SWIR). (a) February. (b) April. (c) October. (d) November.

reflectance increased. The reflectance of the off-year bamboo forest decreased in the near-infrared band during the period of leaf replacement. Because of the reflectance differences near the red-edge and near-infrared regions between on- and off-year bamboo forests, bamboo phenological characteristic index (BPCI) was constructed to differentiate them

$$\mathrm{BPCI} = \frac{\left(\mathrm{Band_j}/\mathrm{Band_k} - \mathrm{Band_8}/\mathrm{Band_i}\right)}{\left(\mathrm{Band_j}/\mathrm{Band_k} + \mathrm{Band_8}/\mathrm{Band_i}\right)} \tag{2}$$

where i denotes 11 or 12, representing the SWIR-1 or SWIR-2 band on the Sentinel-2 image; j is 4 or 5, representing the red or red-edge 1 band on the Sentinel-2 image; and k is 7 or 8, representing the red-edge 3 or near-infrared band on the Sentinel-2 image.

The seasonal spectral variations of moso bamboo canopy can provide valuable information for its extraction, especially in the range from red to near-infrared bands, i.e., band 4–8 of Sentinel-2. The differences between two remote sensing images acquired in different seasons were calculated to illustrate the seasonal variations in canopy reflectance of different forest types, respectively (Fig. 5). It can be observed that coniferous forests, broadleaf forests, and moso bamboo forests exhibit noticeable different seasonality of reflectance across different bands. Indices in ratio form between long-wave bands (band 7 or band 8) and short-wave bands (band 4 or band 5) can effectively differentiate nonbamboo forests and bamboo forests with different phenological statuses. The reflectance seasonality of on-year bamboo forests is the biggest while that of off-year bamboo forests is the smallest. Therefore, BPCI-2 was constructed as follows:

$$\mathrm{BPCI-2} = \frac{\left(\mathrm{Band_{m(j)}}/\mathrm{Band_{n(j)}} - \mathrm{Band_{m(i)}}/\mathrm{Band_{n(i)}}\right)}{\left(\mathrm{Band_{m(j)}}/\mathrm{Band_{n(j)}} + \mathrm{Band_{m(i)}}/\mathrm{Band_{n(i)}}\right)} \tag{3}$$

where i is 4 or 5, representing the red or red-edge 1 band on the Sentinel-2 image; j is 7 or 8, representing the red-edge 3 or near-infrared band on the Sentinel-2 image; and m and n are the imaging months of Sentinel-2 data. m denotes October or November and n denotes February or April here.

To determine the optimal band combination to construct MBI, BPCI, and BPCI-2, the linear discriminant analysis (LDA) [43] and tenfold cross validation were applied to evaluate the classification ability of each alternative indices for different forest types samples obtained from field inventory. LDA constructs linear equations based on the principle of

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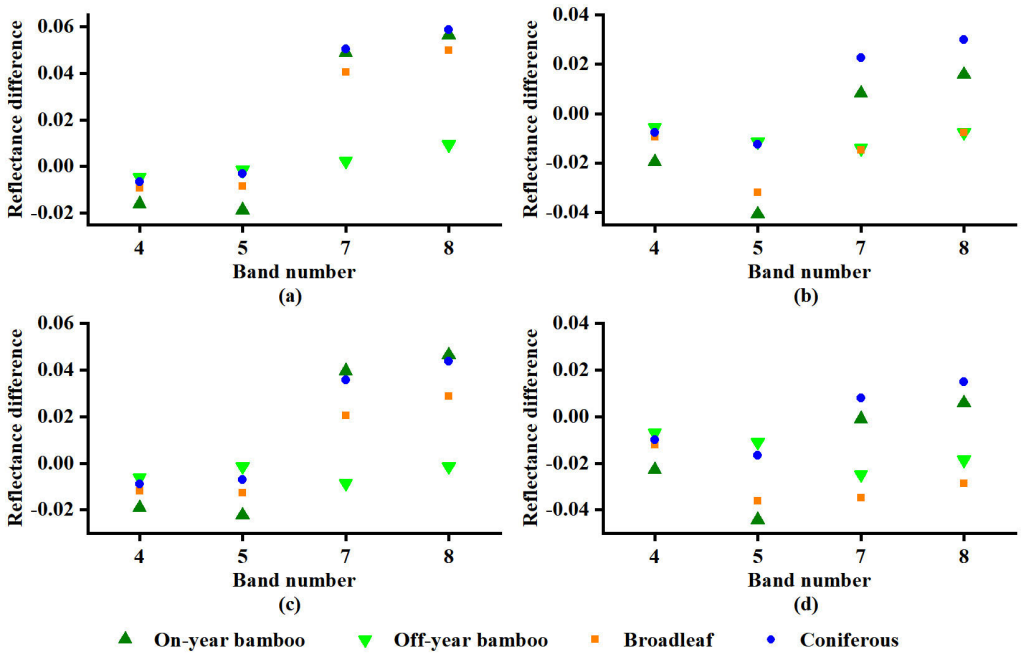


Fig. 5. Reflectance difference of each forest type. (a) October–February. (b) October–April. (c) November–February. (d) November–April.

maximizing the between-class distance and minimizing the within-class distance. The higher LDA score represents the greater potential of indices to distinguish bamboo forests.

D. Synthesizing Time Series Data for the Proposed Indices

Like traditional GANs, CycleGAN also consists of generators and discriminators [38]. The generator is used for image transformation, while the discriminator is used to determine the transformation result. The loss function of CycleGAN consists of four parts, including two adversarial losses and two cycle consistency losses. The adversarial loss function formula of the transformation from the X domain to the Y domain is

$$\mathcal{L}_{GAN}(G, D_Y, X, Y) = \mathbb{E}_{y\sim p_{data}(y)}\left[\log D_Y(y)\right] + \mathbb{E}_{x\sim p_{data}(x)}\left[\log(1 - D_Y(x))\right] \quad (4)$$

where G represents the generator used to transform image from the X domain into image resembling those from the Y domain. D_Y denotes the discriminator used to distinguish the generated image $G(x)$ and the real image from the Y domain. The adversarial loss function for the transformation from the Y domain to the X domain is vice versa.

Cycle consistency is the core of CycleGAN, which ensures that images can be transformed from one domain to another while preserving the integrity of the original image information. Its loss function is

$$\mathcal{L}_{cyc}(G, F) = \mathbb{E}_{x\sim p_{data}(x)}\left[\|F(G(x)) - x\|_1\right] + \mathbb{E}_{y\sim p_{data}(y)}\left[\|F(G(y)) - y\|_1\right]. \quad (5)$$

A total of 1200 training images were cropped from the existing good-quality optical and SAR remote sensing data using a random windowing approach. The size of each training image was 500 × 500. CycleGAN was run by setting the epochs as 100, batch size as 1, and learning rate (LR) as 0.0002.

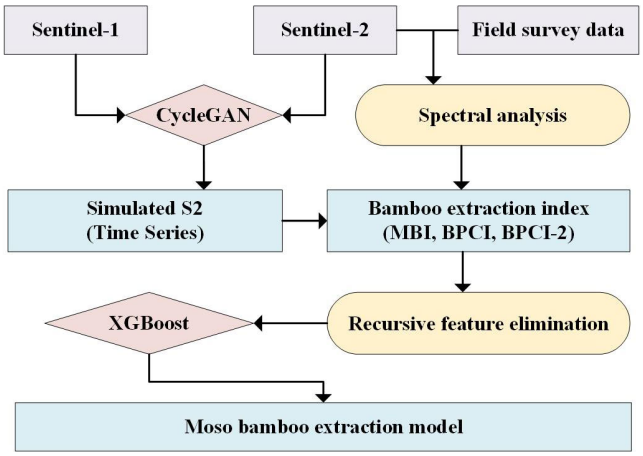


Fig. 6. Framework to map moso bamboo forest.

According to the formulas of proposed indices, simulations were performed for bands 4–9 and 11–12 of Sentinel-2 data. The input optical training data was divided into three batches, i.e., establishing the mapping relationships of two backscattering coefficients of Sentinel-1 with optical images for bands 4–6, bands 7–9, and bands 11–12, respectively, to reduce training costs. Using the synthesized Sentinel-2 images to generate the proposed three indices to fill the gaps under the condition of cloud-free Sentinel-2 data unavailable.

E. Extraction of Bamboo Forest

Fig. 6 illustrates the framework of bamboo forests extraction processes, which primarily consist of the following steps: 1) data collection and preprocessing; 2) spectral analysis of on- and off-year moso bamboo canopies to create novel

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TABLE I
THREE EXISTING INDICES USED IN THIS STUDY

Index	Formula	Reference
Bamboo Index (BI)	$\frac{\text{NDVI-NDWI}}{\text{NDVI+NDWI}}$	[21]
Yearly Change Bamboo Index (YCBI)	$\frac{\text{Band}_7(2020_m)+\text{Band}_8(2020_m)+\text{Band}_9(2020_m)}{\text{Band}_7(2021_m)+\text{Band}_8(2021_m)+\text{Band}_9(2021_m)}$	[24]
Monthly Change Bamboo Index (MCBI)	$\frac{(\text{Band}_7(\text{Apr})+\text{Band}_8(\text{Apr})+\text{Band}_9(\text{Apr}))}{(\text{Band}_7(n)+\text{Band}_8(n)+\text{Band}_9(n))}$	[29]

Note: m denotes Feb or Nov; n denotes Oct or Nov. NDWI denotes the normalized difference water index [49]. The formula of NDVI and NDWI were given in Appendix (Table VI).

TABLE II
DESIGNED EXPERIMENTAL MODELS*

Scenario	Index combination
A	BI_{Feb}, BI_{Apr}, BI_{Oct}, BI_{Nov}
B	$MCBI_{Apr\text{-}Oct}$, $MCBI_{Apr\text{-}Oct}$, $YCBI_{Feb}$, $YCBI_{Nov}$
C	BI_{Feb}, BI_{Apr}, BI_{Oct}, BI_{Nov}, $MCBI_{Apr\text{-}Oct}$, $MCBI_{Apr\text{-}Oct}$, $YCBI_{Feb}$, $YCBI_{Nov}$
D	MBI_{Feb}, MBI_{Apr}, MBI_{Oct}, MBI_{Nov}, $BPCI_{Feb}$, $BPCI_{Apr}$, $BPCI_{Oct}$, $BPCI_{Nov}$, $BPCI\text{-}2_{Nov\text{-}Feb}$, $BPCI\text{-}2_{Nov\text{-}Apr}$, $BPCI\text{-}2_{Oct\text{-}Feb}$, $BPCI\text{-}2_{Oct\text{-}Apr}$
E	$\mathbf{BI_{Jan}}$, BI_{Feb}, $\mathbf{BI_{Mar}}$, BI_{Apr}, $\mathbf{BI_{May}}$, $\mathbf{BI_{Jun}}$, $\mathbf{BI_{Jul}}$, $\mathbf{BI_{Aug}}$, $\mathbf{BI_{Sep}}$, BI_{Oct}, BI_{Nov}, $\mathbf{BI_{Dec}}$
F	$\mathbf{MCBI_{Apr\text{-}May}}$, $MCBI_{Apr\text{-}Oct}$, $MCBI_{Apr\text{-}Oct}$, $YCBI_{Feb}$, $\mathbf{YCBI_{May}}$, $YCBI_{Nov}$
G	$\mathbf{BI_{Jan}}$, BI_{Feb}, $\mathbf{BI_{Mar}}$, BI_{Apr}, $\mathbf{BI_{May}}$, $\mathbf{BI_{Jun}}$, $\mathbf{BI_{Jul}}$, $\mathbf{BI_{Aug}}$, $\mathbf{BI_{Sep}}$, BI_{Oct}, BI_{Nov}, $\mathbf{BI_{Dec}}$, $\mathbf{MCBI_{Apr\text{-}May}}$, $MCBI_{Apr\text{-}Oct}$, $MCBI_{Apr\text{-}Oct}$, $YCBI_{Feb}$, $\mathbf{YCBI_{May}}$, $YCBI_{Nov}$
H	$\mathbf{MBI_{Jan}}$, MBI_{Feb}, $\mathbf{MBI_{Mar}}$, MBI_{Apr}, $\mathbf{MBI_{May}}$, $\mathbf{MBI_{Jun}}$, $\mathbf{MBI_{Jul}}$, $\mathbf{MBI_{Aug}}$, $\mathbf{MBI_{Sep}}$, MBI_{Oct}, MBI_{Nov}, $\mathbf{MBI_{Dec}}$, $\mathbf{BPCI_{Jan}}$, $BPCI_{Feb}$, $\mathbf{BPCI_{Mar}}$, $BPCI_{Apr}$, $\mathbf{BPCI_{May}}$, $\mathbf{BPCI_{Jun}}$, $\mathbf{BPCI_{Jul}}$, $\mathbf{BPCI_{Aug}}$, $\mathbf{BPCI_{Sep}}$, $BPCI_{Oct}$, $BPCI_{Nov}$, $\mathbf{BPCI_{Dec}}$, $\mathbf{BPCI\text{-}2_{Dec\text{-}Jan}}$, $BPCI\text{-}2_{Nov\text{-}Feb}$, $BPCI\text{-}2_{Nov\text{-}Apr}$, $BPCI\text{-}2_{Oct\text{-}Feb}$, $BPCI\text{-}2_{Oct\text{-}Apr}$

*In Scenarios A, B, C, and D, only features were calculated using real Sentinel-2 imageries. In Scnarios E, F, G, and H, features calculated from both real and synthesized Sentinel-2 imageries were included. Features calculated using synthesized imageries are in black font.

bamboo forest extraction indices; 3) synthesis of optical and SAR images using CycleGAN to generate the time series of proposed indices; and 4) construction of bamboo forest extraction models using recursive feature elimination (RFE) [44] and extreme gradient boosting (XGBoost) [45]. The construction process of bamboo extraction model, after data collection and preprocessing, was carried out using Python 3.9.12.

Since the canopy spectra of moso bamboo forests vary notably during different phenological stages, it is necessary to preliminarily extract the forest area to eliminate the impact of human activities and nonforest area changes (e.g., clear-cutting, artificial construction) on the final identification results. Using real and synthesized Sentinel-2 reflectance and normalized difference vegetation index (NDVI) [46] as identification variables, the forest information of study area was extracted by XGBoost classifier. This algorithm, like the random forest, is a tree-based integration algorithm. The difference between these two methods is that the random forest establishes an integrated classifier containing multiple decision trees by bagging, and determines the output category of samples by voting. XGBoost determines the sample category by continuously adding a new base classifier to fit the residual of the error function from the previous model to the predicted value. Previous studies have demonstrated that XGBoost outperformed random forest in most cases [47], [48].

Then, the three indices proposed in this study were used to separate bamboo from broadleaf and coniferous forests. For comparison, three most representative bamboo extraction indices were also used (Table I). A total of eight experimental scenarios were designed (Table II). The candidate variables in the first four scenarios were calculated from real optical imageries. Scenarios A and B were designed to evaluate the bamboo recognition ability of BI and the combination of YCBI and MCBI, respectively. Scenario C was used to assess the bamboo extraction performance with the collaboration of BI, MCBI, and YCBI. Scenario D was to investigate the ability of the model integrating the three proposed indices to map bamboo forest. In the remaining four scenarios, the indices calculated from both real and synthesized imageries were used to investigate whether the synthesis of optical and SAR data was able to better bamboo forest mapping.

The sample data was randomly divided into training set and test set at a 7:3 ratio. The training set was used for feature selection and model construction. The optimal feature combination for each scenario was determined by RFE. The core of RFE is to specify a base evaluator (the XGBoost was applied here) that can provide feature importance information, and

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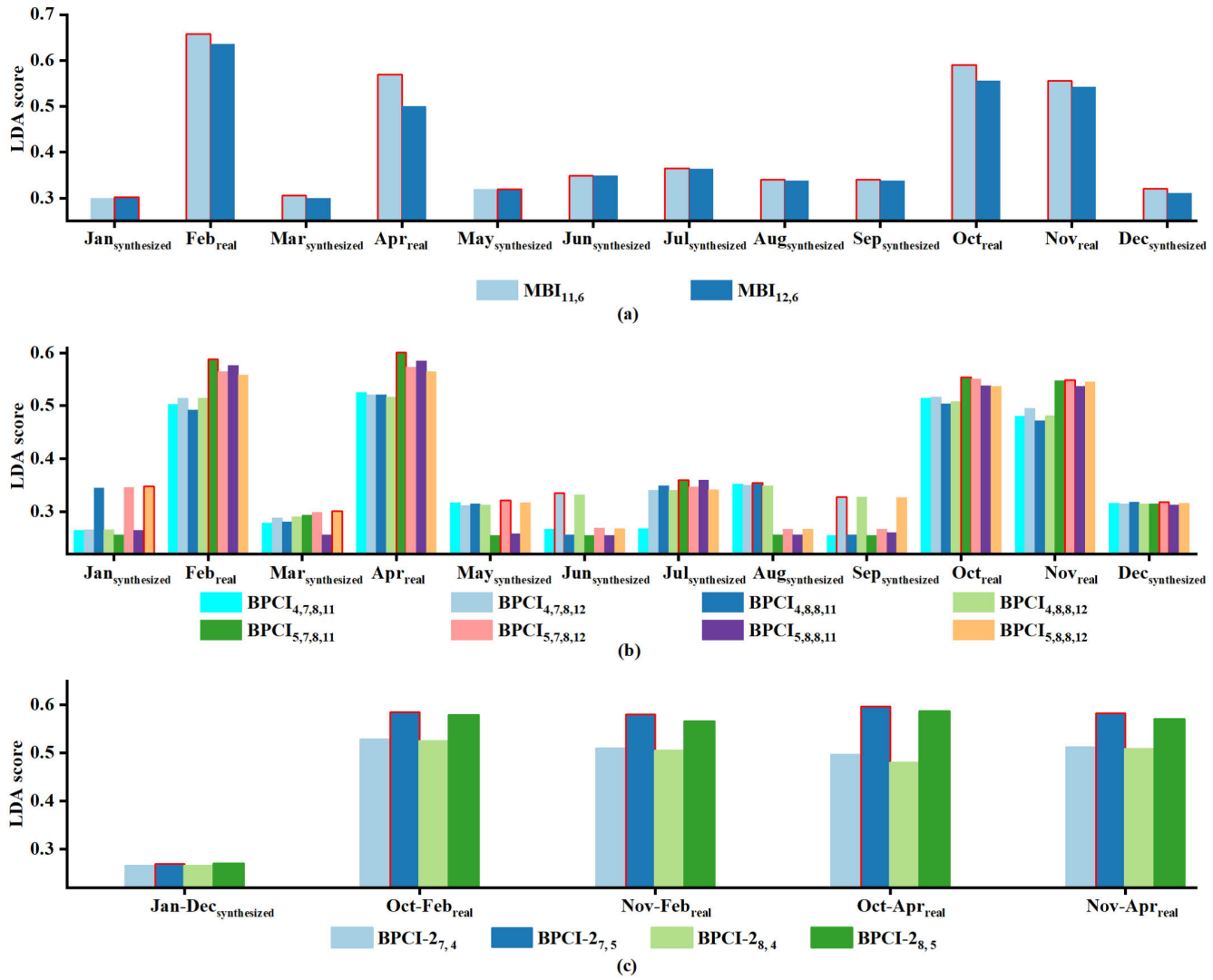


Fig. 7. LDA screened results of (a) MBI, (b) BPCI, and (c) BPCI-2. The indices with the highest LDA scores within the compared group were indicated by the red border.

then iteratively eliminate the features with the weakest contribution in the current candidate set until the model performance is no longer obviously improved (the accuracy improvement amplitude is less than 1%). Since the RFE screening result has slightly different when input different data (i.e., the same training set but with different sample orders) during the iteration process, we set a rule here to determine the final features. First, the training data was shuffled ten times and ran RFE separately. Then the frequency of selection for each candidate was recorded, and candidates selected eight times or more were considered as input features for that specific scenario.

After determining the optimal feature combination for each scenario, the XGBoost was used as the classifier to construct the bamboo forest extraction model. The values of main hyperparameters, i.e., LR, n_estimators (NT), max_depth (MD), min_child_weight (MCW), gamma (GAM), subsample (SS), and colsample_bytree (CB), in XGBoost were determined through grid search. Then, cross validation was performed to evaluate the training effect of the model. Subsequently, the test set data was used to assess the robustness of the constructed model. The confusion matrix was established based on the test set result. Overall accuracy (OA), user's accuracy (UA), and producer's accuracy (PA) were calculated for evaluating model performance.

III. Results

A. Feature Selection

The optimal band combination for constructing MBI, BPCI, and BPCI-2 was determined in each time phase based on the LDA score (Fig. 7).

For real imageries, bands 11 and 6 were more suitable for constructing MBI. For BPCI, the combination of bands 5, 7, 8, and 11 performed the best in February, April, and October. While in November, the combination of bands 5, 7, 8, and 12 outperformed other combinations. As for BPCI-2, the combination of bands 7 and 5 achieved the best performance. For synthesized imageries, bands 11 and 6 were mostly the optimal combination for MBI, except for MBI_{Jan} and MBI_{May}. The best band combination for BPCI varied across different time periods, with bands 5, 8, 8, and 12 for $BPCI_{Jan}$ and $BPCI_{Mar}$, and bands 5, 7, 8, and 12 for $BPCI_{May}$ and $BPCI_{Dec}$. The optimal band combination of $BPCI_{Jun}$ and $BPCI_{Sep}$ was

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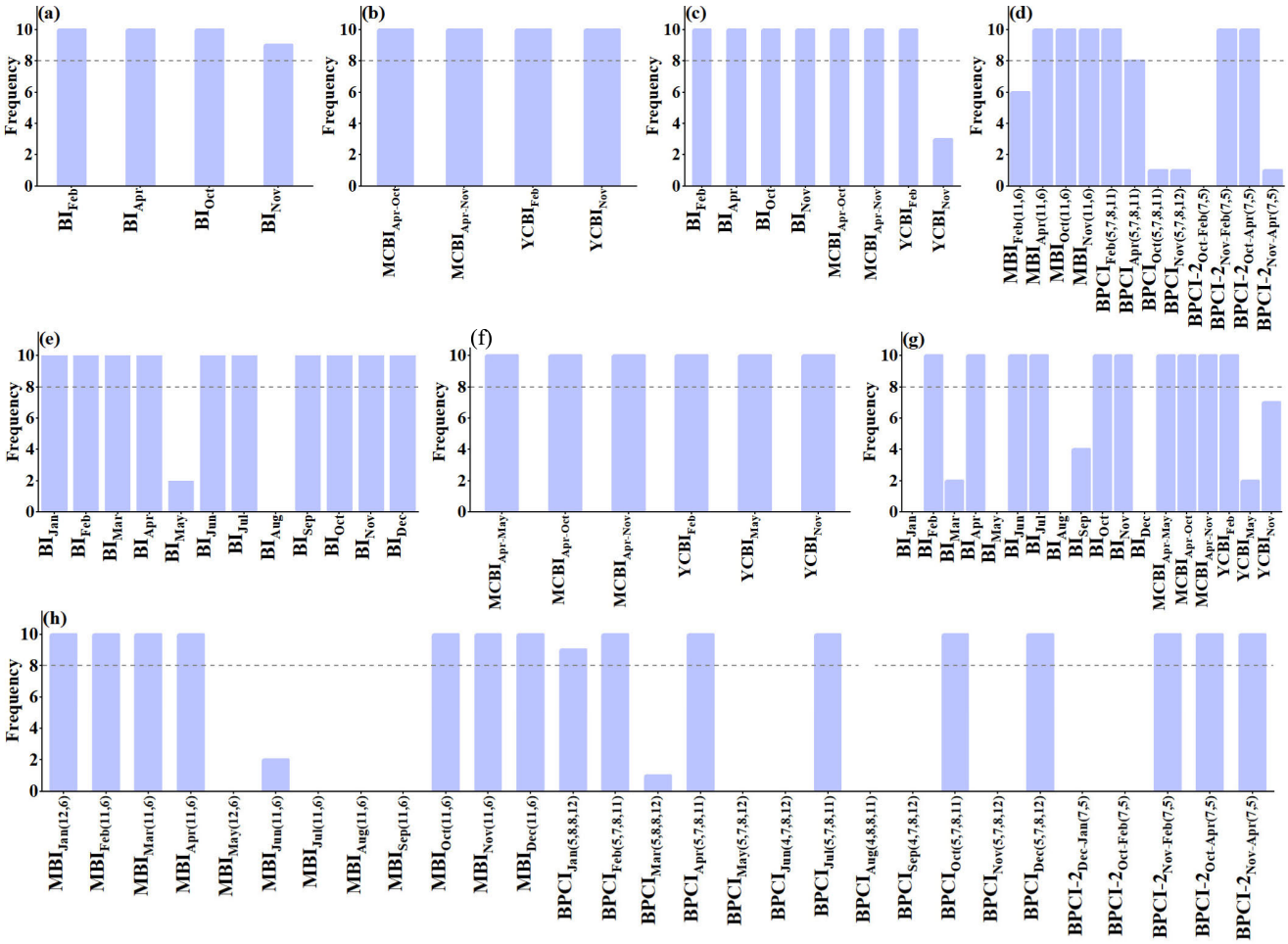


Fig. 8. RFE screened results of each model. (a)–(h) Scenarios A–H.

bands 4, 7, 8, and 12, while bands 5, 7, 8, and 11 and bands 4, 8, 8, and 11 were more suitable for $BPCI_{Jul}$ and $BPCI_{Aug}$, respectively. The optimal band combination for BPCI-2 was the same for the synthesized and real images.

The optimal feature combination for each scenario was determined using RFE (Fig. 8). In Scenarios A and B, all candidate features were selected as inputs. As for Scenario C, all candidate features were selected except for $YCBI_{Nov}$. For Scenario D, seven features were chosen. After incorporating indices calculated from synthesized images, ten features were selected in Scenario E. In Scenario F, all candidates were selected as input features. In Scenario G, all BI, YCBI, and MCBI were used as the candidates and ten features were selected, including six BI features, three MCBI features, and one YCBI features. As for Scenario H, a total of 16 candidates were selected as inputs, including all seven features selected in Scenario D, three features calculated from real imageries (MBI_{Feb}, $BPCI_{Oct}$, and $BPCI\text{-}2_{Nov-Apr}$), and six features (MBI_{Jan}, MBI_{Mar}, MBI_{Dec}, $BPCI_{Jan}$, $BPCI_{Jul}$, and $BPCI_{Dec}$) calculated from synthesized imageries.

B. Extraction Accuracy of Bamboo Forests

Table III shows OA values and calibrated hyperparameters of individual scenarios. For all models, OA vales of the training and test sets are almost identical, indicating that there is no overfitting problem during the training process and trained models were robust.

When real Sentinel-2 images were used (i.e., Scenarios A–D), the OA values of Scenario A exceeded 81% for both training set and test set. The performance of Scenario B was relatively poor with OA vales below 70% for both training set and test set. The collaboration of BI, YCBI, and MCBI (Scenario C) improved the accuracy of identifying bamboo forests, with OA equal to 86.54 $\pm$ 0.62% for the training set and 86.61% for the test set. The model based on the three indices proposed in this study (Scenario D) performed the best, with OA equal to 88.25 $\pm$ 0.58% for the training set and 87.99% for the test set.

The addition of features calculated from synthesized imageries improved model performance. Scenarios E and F, which included the features used in Scenarios A and B, and features calculated from synthesized imageries, achieved noticeable improvement in OA for both the training set and test set. The OA values for the training and test sets in Scenario F were 8.23% and 9.5% higher than the corresponding values in Scenario B. The OA values in Scenario G were 5.28% and 3.78% higher than the corresponding values in Scenario C for the training set and test set. In Scenario H, MBI, BPCI, and BPCI-2 calculated from both real and synthesized imageries were used as input features. The OA values were 93.49 $\pm$

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TABLE III
CALIBRATED HYPERPARAMETERS AND ACCURACY OF INDIVIDUAL MODELS

Scenario	Hyper-parameters							OA (%)	
	LR	NT	MD	MCW	GAM	SS	CB	Training-set	Test-set
A	0.04	50	5	1	0	0.8	1	81.10 ± 0.53	81.12
B	0.3	52	6	1	0	0.8	1	66.21 ± 0.76	63.73
C	0.3	100	6	1	0	1	0.9	86.54 ± 0.62	86.61
D	0.3	92	6	1	0	0.7	0.4	88.25 ± 0.58	87.99
E	0.3	112	6	1	0	0.8	0.8	88.59 ± 0.54	88.67
F	0.44	130	6	1	0	0.7	1	74.44 ± 0.72	73.23
G	0.3	79	6	1	0	0.8	0.9	91.82 ± 0.57	90.39
H	0.3	151	6	1	0	0.9	0.8	93.49 ± 0.52	94.97

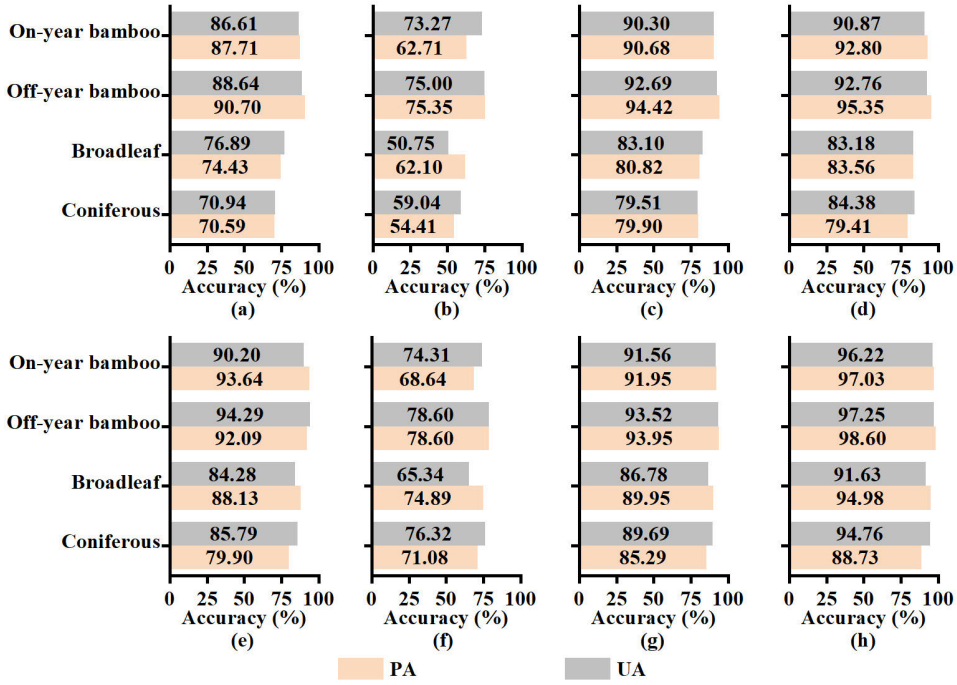


Fig. 9. User's (UA) and producer's (PA) accuracy of the test set. (a)–(h) Scenarios A–H.

0.52% and 94.97% for the training and test sets, 5.24% and 6.98% higher than the corresponding values in Scenario D, in which only the feature calculated from real imageries were used.

The ability of individual models to separate different forest types for the test set was further examined (Fig. 9). The poor recognition accuracy of Scenarios A and B for broadleaf and coniferous forests is the main reason for their low OA values. However, their ability to classify bamboo forests was acceptable. For on-year bamboo samples, the UA and PA values of Scenario A were 86.61% and 87.71%, respectively. For off-year bamboo samples, both UA and PA of Scenario A exceeded 88%. The combination of YCBI and MCBI (Scenario B) performed relatively poor. The UA and PA values for on- and off-year bamboo samples were below 76%. In comparison with Scenarios A and B, the ability of Scenario C to distinguish four types of samples was notably improved with the UA and PA values for both on- and off-year samples above 90%. The collaboration of MBI, BPCI, and BPCI-2 (Scenario D) performed better for identifying on- and off-year bamboo samples than that of existing bamboo extraction indices.

As for the model driven by features from real and synthesized imageries, the performance of the model integrating YCBI and MCBI (Scenario F) was still unsatisfactory. Although the accuracy for nonbamboo samples was greatly improved relative to Scenario B, the PA and UA values for bamboo samples were below 80%. In Scenario E, the model with BI time series data as the input features performed well for both on- and off-year samples, with PA and UA values comparable to those in Scenario G. In Scenario G, the model performed better for nonbamboo samples. The model in Scenario H outperformed all other models for all types of samples, achieving UA values of 96.22% and 97.25%, and PA values of 97.03% and 98.60% for on- and off-year samples, respectively.

Fig. 10 shows the classification confusion matrix of individual models. The model of Scenario A misclassified 4.70% of coniferous and broadleaf samples as bamboo samples. The misclassification rate of bamboo samples was only 2.50%. The highest misclassification rate was observed for coniferous samples at 6.37%. About 76.92% of the misclassified coniferous samples were mistakenly recognized as bamboo samples. As for the model of Scenario B, 18.16%

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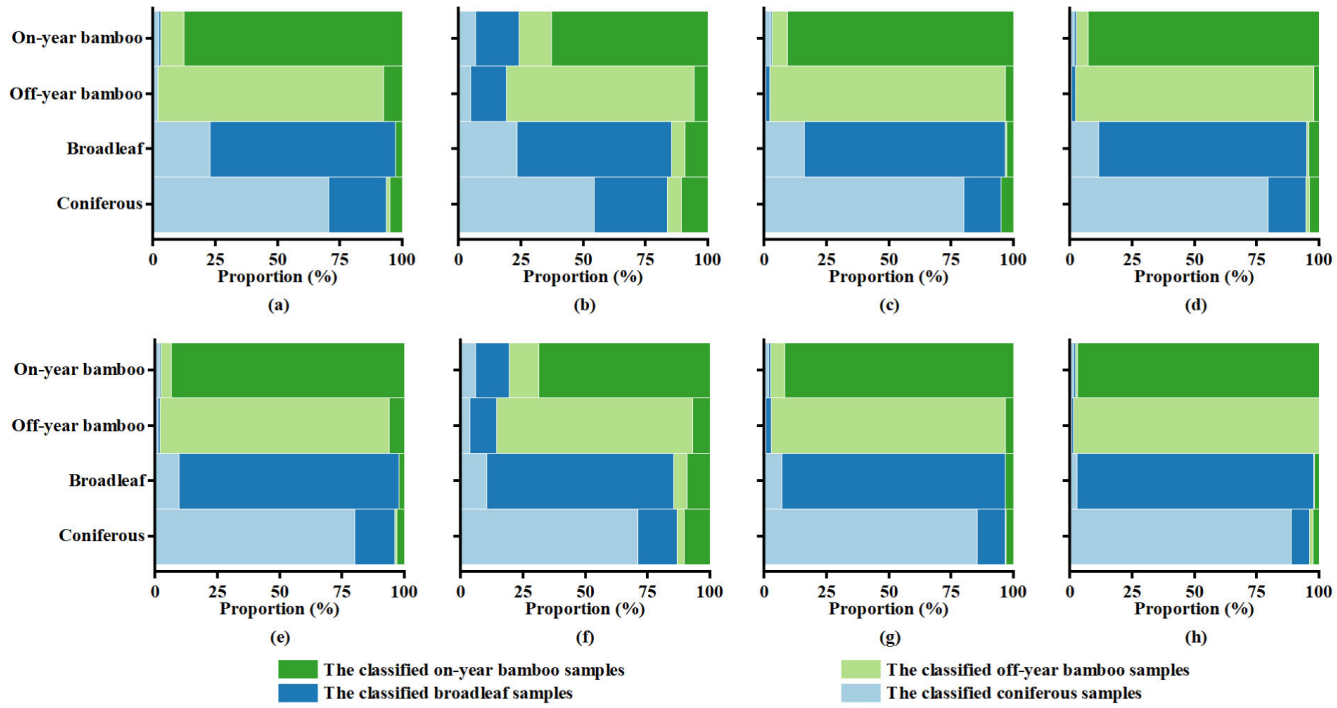


Fig. 10. Confusion matrix of individual models. (a)–(h) Scenarios A–H.

of nonbamboo samples and 27.76% of bamboo samples were misclassified, respectively. The model of Scenario C outperformed the models of both Scenarios A and B, with a reduced misclassification rate of 4.19% for nonbamboo samples, while the misclassification rate of bamboo samples was slightly higher than that of Scenario A (2.73%). The model of Scenario D had a misclassification rate of 5.49% for nonbamboo samples, 1.3% higher than that of Scenario C, mainly due to the misclassification of broadleaf samples, with 81.82% of misclassified samples assigned to the on-year bamboo samples. The addition of features calculated from synthesized imageries reduced the sample misclassification rate for each scenario. The results revealed that in most scenarios, the misclassified nonbamboo samples were incorrectly categorized as on-year bamboo samples. The model of Scenario H exhibited the best performance, with a misclassification rate of only 1.81% for bamboo samples and 3.17% for nonbamboo samples.

C. Distribution of On- and Off-Year Bamboo

Overall, the model with MBI, BPCI, and BPCI-2 from real and synthesized imageries as inputs (Scenario H) performed the best, followed by the model in Scenario G. Therefore, the bamboo distribution information in the study area was extracted using models of these two scenarios, respectively. The extracted area of on-year bamboo, off-year bamboo, and total bamboo forest was calculated (Table IV).

It can be observed that the extracted areas of bamboo forest using the models of Scenarios G and H were 397.01 and 444.78 km^2, respectively. The area of off-year bamboo forest extracted by the model of Scenario G was noticeably less than that from Scenario H. According to official inventory data (http://www.fjsc.gov.cn/cms/html/scxrmzf/2019-03-21/2062313157.html), the total bamboo forest area in Shunchang County was approximately 440 km^2 in 2017. The area of bamboo forest extracted using the model of Scenario H was closer to the official statistical data than that of Scenario G.

TABLE IV
EXTRACTED BAMBOO FOREST AREA OF MODELS

Scenario	Area / km^2		
	On-year bamboo	Off year bamboo	Moso bamboo
G	250.16	146.85	397.01
H	257.00	187.72	444.72

Therefore, the bamboo distribution map of study area in 2020 was produced using the model of Scenario H (Fig. 11). In this map, on- and off-year bamboo was differentiated. Overall, the spatial pattern of extracted bamboo mirrored forestry inventory data surveyed in 2012 well. Nevertheless, there are still some differences between our produced map and inventory data, especially in the boundaries of the inventory polygons. Such differences might be attributed to the uncertainties in both the extracted map and the inventory data. In each polygon of the forest inventory data, only the dominant tree species was recorded. In some polygons with complex vegetation composition, several miscellaneous trees might coexist. In addition, forest inventory was very costly and only implemented every five years. Moso bamboo is an invasive growth vegetation, its distribution might be expanded in five years, especially in the transition boundary between bamboo and other forests.

The bamboo forest in the south, north, and west parts of the study area is mostly distributed in the boundaries of the county, while the bamboo forest in the east part extends toward the county center. According to the terrain conditions of the bamboo forest habitat, the moso bamboo usually grows

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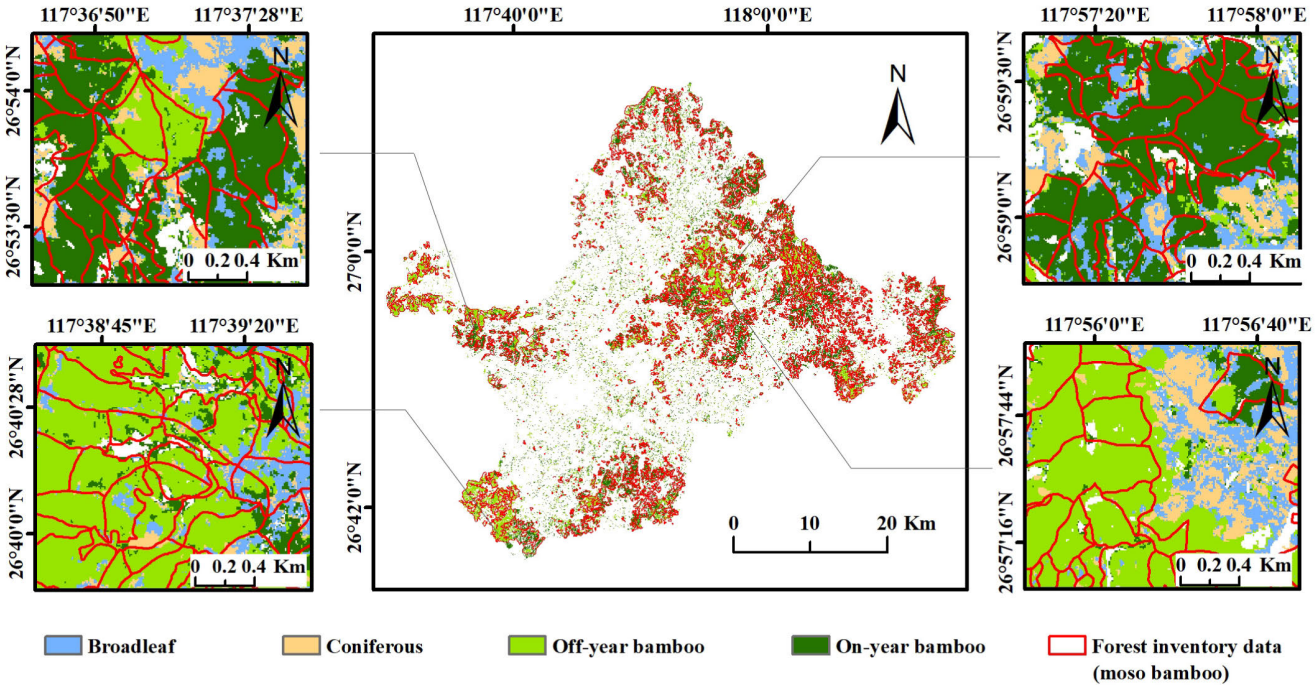


Fig. 11. Spatial consistency between mapped bamboo forest and forest inventory data.

in mountainous areas at an altitude of 200–800 m. The distribution altitude of the on-year bamboo forest in that year was lower than that of the off-year bamboo forest. The slope of the bamboo forest habitat was generally concentrated at 10°–40°. Bamboo was mostly distributed at slopes from 20° to 25°. The aspect of bamboo forest habitat was relatively close to the northwest and southeast directions (Fig. 12).

IV. Discussion

A. Sensitivity of Individual Indices to Topography

Among Scenarios A–D, the model of Scenario D integrating MBI, BPCI, and BPCI-2 performed the best. One possible reason for this phenomenon is the different sensitivity of individual indices. The calculation formula of the proposed indices allows them to mitigate the influence of topography to some extent. This merit is crucial for mapping bamboo forest in mountainous region, such as the study area here.

The features used in Scenarios C and D were selected to analyze the impacts of topography on individual indices, which were assessed by comparing their changes with aspects and slopes (Fig. 13). Pixels in eight slope ranges (i.e., slope $<$ 10°, 10° $\leq$ slope $<$ 15°, 15° $\leq$ slope $<$ 20°, 20° $\leq$ slope $<$ 25°, 25° $\leq$ slope $<$ 30°, 30° $\leq$ slope $<$ 35°, 35° $\leq$ slope $<$ 40°, slope $\geq$ 40°) were first grouped. The indices of pixels corresponding to different aspects in individual slope groups were then averaged.

MBI, BPCI, and BPCI-2 were constructed in the normalized ratio forms. Therefore, they showed low sensitivity to topography. The means of MBI, BPCI, and BPCI-2 changed marginally with aspects and slopes. BI is constructed on the basis of two ratio-type indices (i.e., NDVI and NDWI), so the influence of topography on this index was also relatively small. However, in steep areas (especially in areas with slopes above

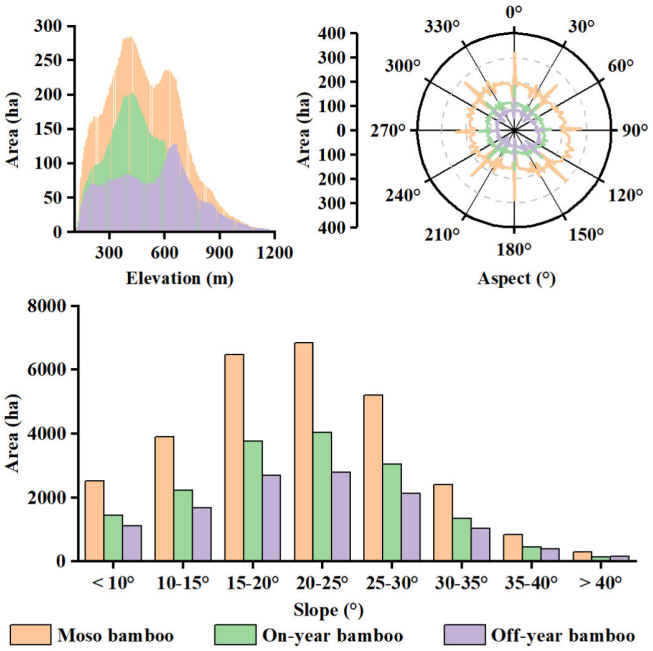


Fig. 12. Changes of bamboo forest area with elevations, slopes, and aspects.

40°), the values of MBI, BPCI, and BI differed observably between shaded and sunny aspects.

YCBI is calculated using data in the same month of two adjacent years. BPCI-2 and MCBI are calculated using data in different months of the same year. These three indices are in ratio forms. Conceptually, they are the same in the sensitivity to topography. However, our data shows that YCBI and MCBI are more sensitive to topography than BPCI-2. YCBI and MCBI were high at shaded aspects and low at sunny aspects. Their changes with aspects are more obvious at steep

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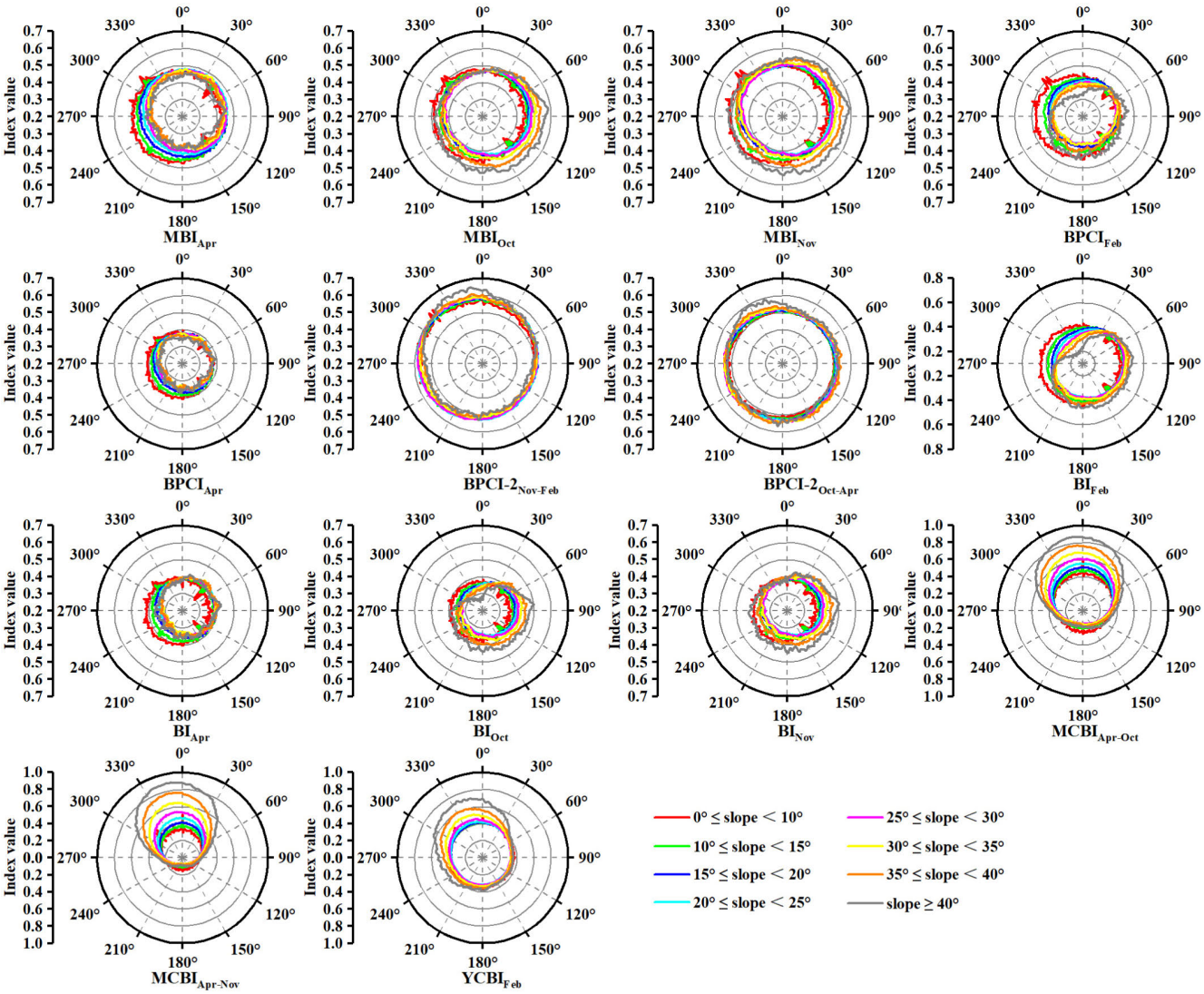


Fig. 13. Changes of mean indices from real imagery with slopes and aspects. The polar angle denotes aspects, while the radius denotes the mean value of index.

locations. The main reason might be attributed to changes in sun angles and soil wetness between different years and months.

Constrained by weather condition, we were unable to calculate MCBI from real imageries using the original formula, i.e., using the ratio of the summation of reflectance in bands 7–9 in April to that in May [29]. Hence, we approximated the calculation of MCBI by combining imageries from April and either October or November. As shown in Table V, the average solar zenith and azimuth angles in April were noticeably smaller than those in October and November. The reflectance in April showed much smaller change with aspects than that in October or November [Fig. 14(a)], resulting in the high sensitivity of MCBI to topography. Therefore, if imageries acquired in two months with large differences in sun angles were used, the sensitivity of this index to topography should be considered.

The influences of changing sun angles on YCBI are marginal since it is calculated using data in the same month of adjacent years. However, water content greatly affects the observed reflectance of the canopy (the higher the water content, the lower the reflectance of the canopy) [50]. The data measured at local meteorological station indicated that the amount of precipitation between 2020 and 2021 had notable differences (Table V). Specifically, the precipitation in January and February 2020 was higher than that in 2021. Therefore, the reflectance of the remote sensing image in 2020 was lower than that in 2021 [Fig. 14(b)]. Compared with the sunny slope, the solar radiation energy received by the shady slope area was relatively low. Consequently, the temporal variability of wetness in the shady area was smaller than that in the sunny area [51], [52], which cause the reflectance differences between 2020 and 2021 in bands 7–9 at the sunny aspects were more obvious than at the shady aspects. Therefore, YCBI also exhibits high sensitivity to topography.

BPCI-2 was calculated in the ratio form two times. The effect of changing reflectance with aspects can be constrained. The three proposed indices exhibited low sensitivity to topography, resulting in outperformance of models based on MBI, BPCI, and BPCI-2 over models based on BI, MCBI, and YCBI.

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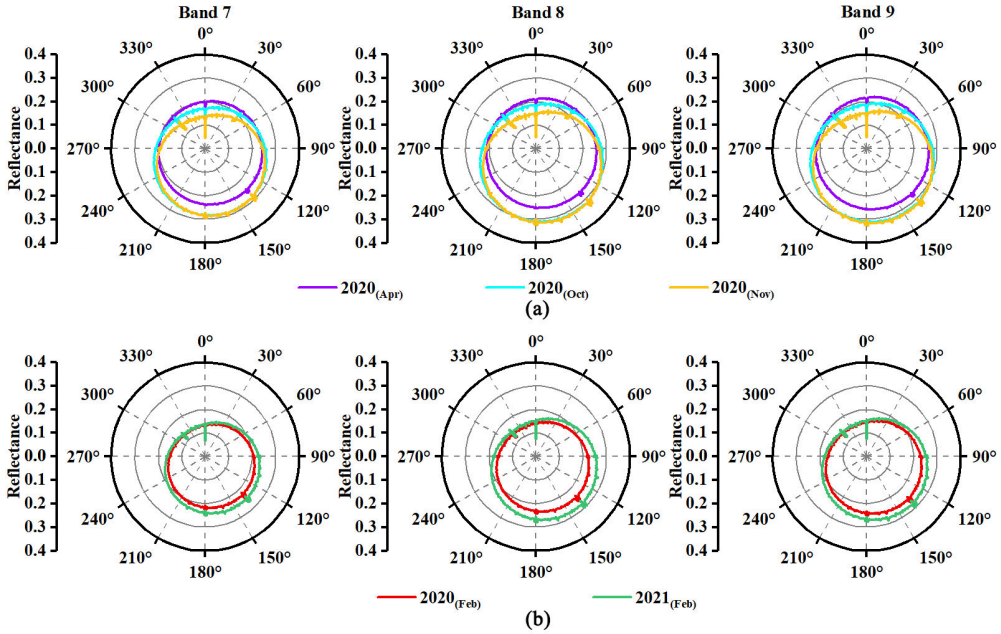


Fig. 14. Comparison of reflectance in different time. (a) Reflectance of bands 7–9 in April, October, and November of 2020 ($2020_{(Apr)}$, $2020_{(Oct)}$, $2020_{(Nov)}$). (b) Reflectance of bands 7–9 in $2020_{(Feb)}$ and $2021_{(Feb)}$.

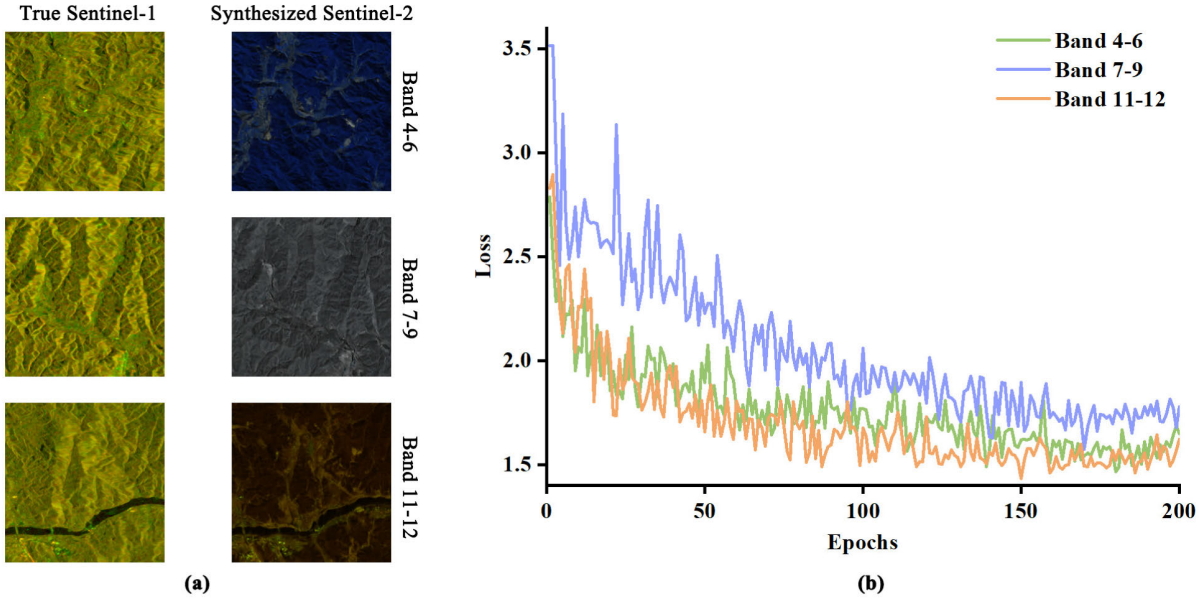


Fig. 15. Synthesized results of CycleGAN. (a) Sentinel-2 images generated from Sentinel-1 imagery using CycleGAN. (b) Train loss of the CycleGAN on Sentinel-2 imagery for bands 4–6, bands 7–9, and bands 11–12, respectively.

B. Role of Synthesizing SAR Data in Bamboo Forest Extraction

Unsupervised domain adaptation has garnered significant attention in the fields of forestry and agriculture remote sensing. CycleGAN, as a representative deep learning algorithm, has the ability to perform image style transfer while preserving the original image's information without paired training data, which has been widely used in image segmentation and data fusion.

Both SAR and optical imagery data inherently provide signals reflecting electromagnetic waves signals from ground objects, with the key distinction being the much longer wavelength of the former. Therefore, the use of CycleGAN for time series data synthesis of optical and microwave data is theoretically feasible. From a visual perspective and the results of loss function iterations, CycleGAN successfully performed style transfer from Sentinel-1 to Sentinel-2 (Fig. 15). The Sentinel-2 imageries in the months of January, May to September, and December were collected. Imagery in March was not collected due to a high cloud cover. The cloud-free portions of each collected image were manually segmented for comparison with the corresponding synthesized images in terms of MBI and BPCI. The $\text{BPCI-2}_{\text{Dec}-\text{Jan}}$ was not calculated because there is no overlap of cloud-free areas in imageries between December and January. It can be observed

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TABLE V
PRECIPITATION AND AZIMUTH ANGLE IN DIFFERENT PERIODS

Year	2020					2021				
Month	Jan	Feb	Apr	Oct	Nov	Jan	Feb	Apr	Oct	Nov
Precipitation / mm	46.21	117.37	109.83	8.38	3.40	4.73	101.77		38.92	163.49
Azimuth angle / °		148.96	132.96	160.04	162.76		149.02			162.24
Zenith angle / °		42.51	23.11	40.32	46.27		42.58			46.21

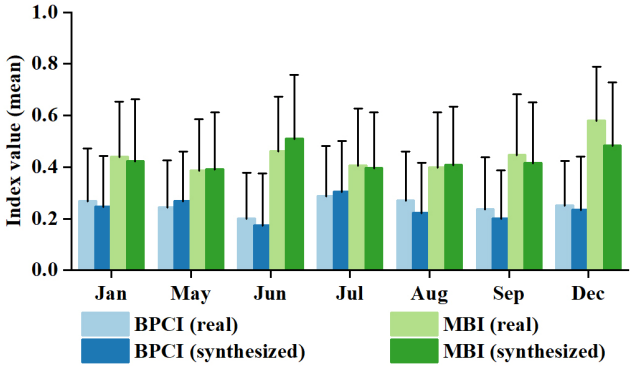


Fig. 16. Comparison of indices calculated from real and synthesized imageries.

that the mean values of the three indices calculated from the synthesized imageries were close to those calculated from the real imageries (Fig. 16), implying the feasibility of synthesizing the optical and SAR imageries to fill optical data gaps for better bamboo forest in areas with abundance of clouds.

It is noteworthy that style transfer from SAR to optical data still encounters uncertainties since CycleGAN only accomplishes style transfer, which involves simulating the data distribution of one style of image onto another style based on the input images. It should be kept in mind that the observation geometry (i.e., the angles of microwave signal transmission and reception) of SAR satellites is consistent across different seasons. However, the temporal changes of solar zenith and azimuth angles might cause seasonal variations of the radiation signals received by optical sensors. Therefore, the distribution of shadows in optical imageries differs from that of SAR data, especially in areas with complex terrains. Meanwhile, there are geometric and phase distortions on SAR imageries in steep areas. Additionally, SAR data is more sensitive to the canopy structure and the dielectric constant of the target, whereas optical imagery primarily provides physicochemical information on the surface properties of forest canopies. Therefore, CycleGAN is unable to fully transform SAR data into realistic optical data. It can learn and incorporate marginal features from optical data to synthesize SAR data to enrich features usable for mapping bamboo forest.

Since there are uncertainties in the synthesized data, the selection of a classification model is particularly crucial. Ensemble algorithms based on tree models are capable of determining weights of individual input features in the training process [45]. This ensures that the model not only learns features from different dimensions but also limits the disturbance of noise in input features.

In this study, the features selected by RFE were used to construct models classifying bamboo forest. The inclusion of features calculated from synthesized imageries improved the accuracy of identifying bamboo forest, indicating that the synthesis of optical and SAR data is a potentially applicable way under the situation without enough cloud-free optical data. The importance of synthesizing SAR and optical data for improving the classification accuracy of bamboo forest in Scenarios G and H was assessed using the shapely additive explanations (SHAP) method [53] (Fig. 17). In Scenario G, the top five important features were calculated from real imageries, including four BI features and one YCBI feature. The importance of $MCBI_{Apr-May}$ calculated from the synthesized imageries was in the sixth place, higher than those of two MCBI features calculated from the real imageries. The importance of two growing-season BI features calculated from the synthesized imageries was the lowest. As to Scenario H, the first six important features were all calculated from real imageries. The importance of $BPCI_{Jul}$, MBI_{Mar}, MBI_{Dec}, $BPCI_{Dec}$, $BPCI_{Jan}$, and MBI_{Jan} calculated from the synthesized imageries only ranked seventh, eighth, tenth, 14th, 15th, and 16th, respectively.

Overall, the features calculated from real imageries played more important roles in identifying bamboo forest than those calculated from the synthesized imageries. Nevertheless, the collaboration of real and synthesized imageries effectively improved the identification accuracy of bamboo forest. This strategy could be suitable for the recognition of other forest types. Of course, how to constrain the uncertainties in the synthesized imageries is worthy of further investigation.

C. Availability and Uncertainty of Proposed Indices

This study proposed MBI, BPCI, and BPCI-2 based on Sentinel-2 data. They demonstrated better ability to identify bamboo forest than indices previously constructed for classifying bamboo forest. However, all proposed indices include the reflectance of red-edge bands, which might be not available in other sensors. For coarse resolution remote sensing data, the presence of mixed pixels can weaken the ability of indices to identify target objects, especially in areas with complex vegetation composition [54]. High spatial resolution imagery can provide clear crown-level outlines, and the associated spectral information is suitable for identifying various types of forests, even tree species. However, such data may only be used for small areas. In contrast, free medium-resolution remote sensing imageries offer a better cost–performance ratio. The introduction of Google Earth Engine (GEE) can notably reduce computational costs for

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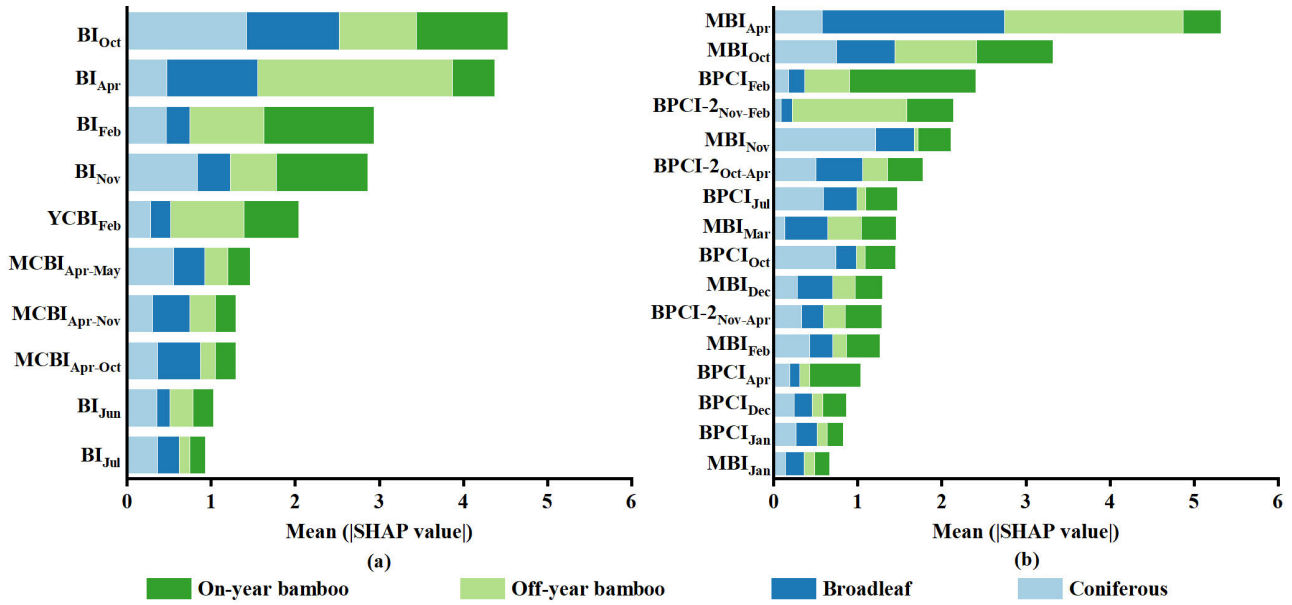


Fig. 17. SHAP evaluation results of (a) Scenario G and (b) Scenario H.

large-scale research [55], [56]. With this powerful platform, it is possible to map bamboo forest at regional and global scales using various indices, such as those proposed in this study.

Moso bamboo forest, as a fast-growing economic forest, is subject to various disturbance events (e.g., selective logging and insect disturbance) every year, which could cause difficulties in bamboo forest mapping. Insect infestation is a type of natural disturbance in forest ecosystems. *Pantana phyllostachysae* Chao is the most threatening local defoliator to the bamboo forest, which has three generations within a year (overwintering generation, the first generation, and the second generation). The damaged period of the overwintering generation began in March, and its threat degree is relatively small [57]. The first generation is active in August–September, which poses the greatest threat to the bamboo forest. The second generation occurs in October, and its threat is less than that of the first generation.

Among the three indices proposed in this study, BPCI and BPCI-2 were primarily used for the phenological state recognition of bamboo forests. BPCI-2 is calculated using red-edge reflectance ratios in early and late months of a year (3). Its performance can be significantly affected by insect disturbances because the activity of second-generation larvae is high in the late months of a year, and the vitality of canopies in many areas severely damaged by the first-generation larvae may not have fully recovered. It can be seen that the separability of BPCI-2 between the on-year bamboo and other tree species in pest damaged area was much lower than that in the noninsect disturbance area. The noninsect disturbance on-year bamboo forest is generally distributed in areas with high values of BPCI-2. While in the pest damaged area, the BPCI-2 value is relatively low (Fig. 18). Since the overwintering larvae have not caused serious damage in February yet, it is necessary to jointly use $BPCI_{Feb}$ for the identification of on- and off-year bamboo, i.e., the analysis results of SHAP also confirmed the significance of $BPCI_{Feb}$ in distinguishing on- and off-year moso bamboo forests. However, precise forestry survey data of bamboo forest insect infestations is mostly lacking. The effective method to detect bamboo forest insect infestations is valuable for mapping bamboo distribution.

Logging is another important disturbance factor in forest ecosystems. The selective logging events reduce the proportion of regional canopy area, while the probability of forest background observed by sensors increases. It can cause dramatic changes in remote signals, e.g., the decline in the indicators that represent canopy greenness and humidity [58]. The logging events mostly occurred in off-year bamboo forests, which is related to the management mode of the moso bamboo forest. To ensure economic benefits, the manager will log the off-year bamboo in most cases instead of on-year bamboo considering the shoot's production in the next year. A previous study found that the greenness and humidity of off-year bamboo leaves were significantly lower than those of on-year bamboo leaves [28]. Theoretically, logging events increase the spectrum difference between on- and off-year bamboo forests. The off-year bamboo forest is mainly located in areas with low values of BPCI and BPCI-2 (Fig. 18). When the logging events occurred, the values of those two indices in that areas were still low. Therefore, the impact of logging on distinguish on- and off-year bamboo forest is relatively small. However, as selective logging activities are widespread in plantation forests, the potential impact of this disturbance factor on the OA of moso bamboo forest identification needs further investigation using higher resolution remote sensing imageries.

This study only assessed the applicability of proposed indices in one region. Nevertheless, theoretical analysis and assessment on the applicability of these indices need further investigation.

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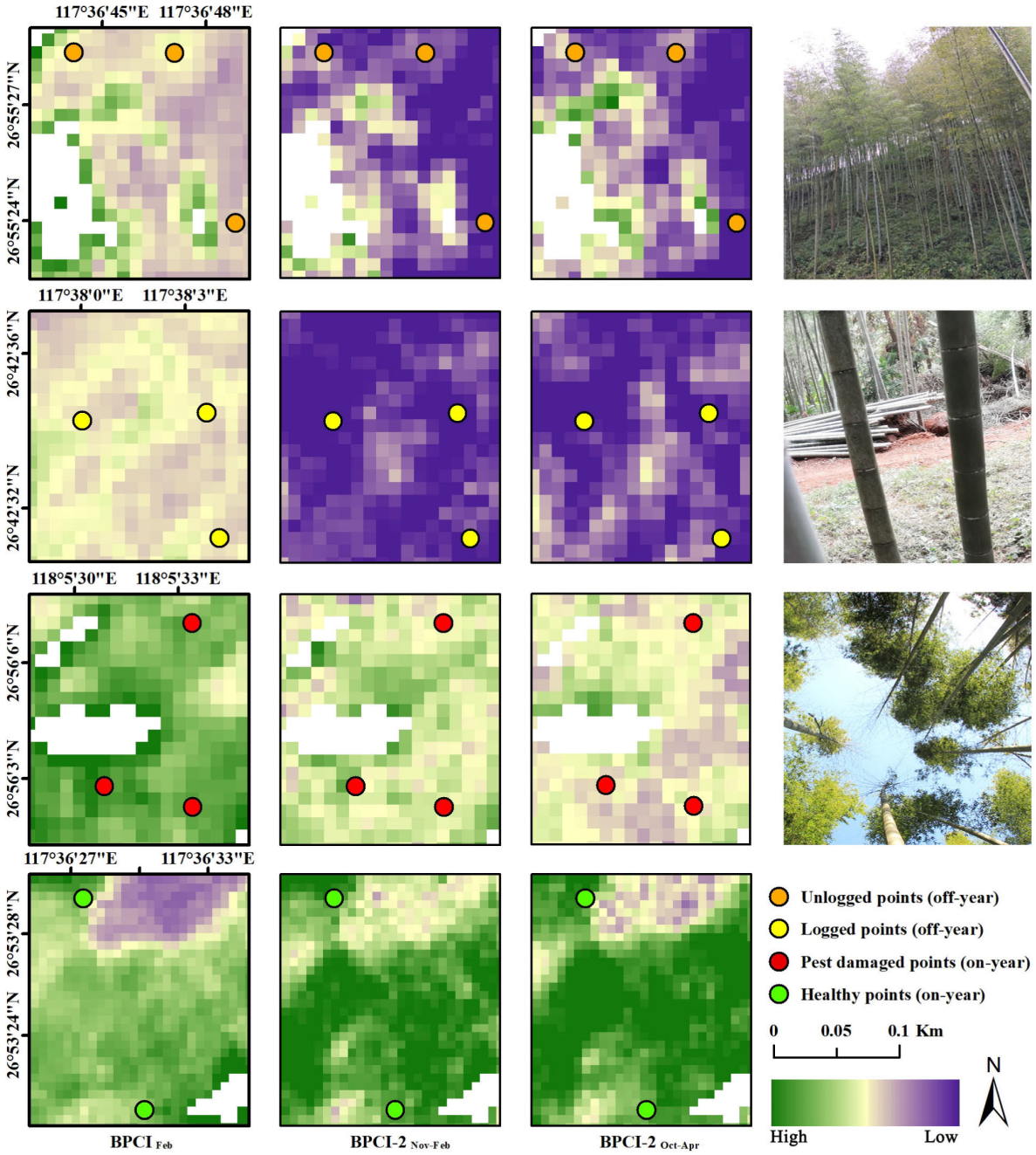


Fig. 18. Impact of forest disturbance on BPCI and BPCI-2. The on-year bamboo forests are mainly distributed in the high-value areas of BPCI and BPCI-2, while off-year bamboo forests are primarily found in the low-value areas of these two indices. However, the value of BPCI-2 in pest damaged areas of on-year bamboo forests decreased notably, while the $BPCI_{Feb}$ value remains high in these regions. The values of $BPCI_{Feb}$ and BPCI-2 differed marginally between the logged and nonlogged areas of off-year bamboo forest.

V. Conclusion

This study proposed three indices (MBI, BPCI, and BPCI-2) for bamboo forest extraction using Sentinel-2 optical imageries. The combination of these three indices was not only effective for extracting bamboo forests but also can be used to identify the phenological states of bamboo forests, i.e., on- and off-year bamboo. The model driven by the three proposed indices outperformed that based on indices previously designed for mapping bamboo forests. One possible reason for the better performance of the proposed indices is their low sensitivity to topography.

The synthesis of optical and SAR data is a potentially applicable way for identifying bamboo forest under the situation without enough cloud-free optical data available. The model driven by indices calculated from both real and synthesized imageries significantly improved the accuracy of differentiating bamboo forests with different phenological statuses. The PA and UA values were 97.03% and 96.22% for on-year bamboo samples, and 98.60% and 97.25% for off-year bamboo samples. They are obviously higher than the corresponding values produced by the model driven by features calculated only from real imageries.

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TABLE VI
CALCULATION FORMULAS OF NDVI AND NDWI

Index	Formula
Normalized Difference Vegetation Index (NDVI)	$\frac{Band_8\text{-}Band_4}{Band_8+Band_4}$
Normalized Difference Water Index (NDWI)	$\frac{Band_{8a}\text{-}Band_{12}}{Band_{8a}+Band_{12}}$

APPENDIX

See Table VI.

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